A Formative Assessment of Traditional Undergraduate Students' Use of an AI Learning Platform in STEM Courses

Abstract

A small to midsized university supported science, technology, engineering, and mathematics (STEM) education by partnering with a reputable online learning company to provide students and faculty access to artificial intelligence powered textbook problem sets and chatbots. An in-house formative assessment study was conducted in Fall 2024 and Spring 2025 to understand how traditional undergraduate students engaged with the platform. Quantitative results suggests that the general population did not utilize it as much as initially expected. Qualitative findings imply that students might have preferred alternative platforms like ChatGPT, though this may have had nothing to do with the company itself. Overall, students sought superficial answers and neglected the conceptual foundations of assignments. They may have been overwhelmed when utilizing these tools given the lack of guidance from instructors on how, if at all, to use them. The same might be true of instructors. One specific recommendation is to increase training opportunities for how to interface with specific AI tools.